Yoonjae Baek

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Education

Pohang University of Science and Technology (POSTECH) — Pohang, South Korea
M.S. in Artificial Intelligence — *Feb 2025 – present*
- Graduate School of Artificial Intelligence, Computer Vision Lab
- Advisor: Prof. Suha Kwak

Sungkyunkwan University (SKKU) — Seoul, South Korea
B.S. in Computer Education — *Mar 2018 – Feb 2025*
- Advisor: Prof. Sungeun Hong

Publications

[1] ***Taming VLAs under Robot Execution Errors: Self-Compensation and Stress Testing***
Sohyun Lee*, **Yoonjae Baek***, Jaesang Won, Jinnyeong Kim, Hyunwoo Kang, Seung-Hwan Baek, Ivan Laptev, Suha Kwak (*equal contribution)
Under Review, 2026

[2] ***When to Switch: Reliable Action-Chunk Extension for Vision-Language-Action Models***
Seonghoon Yu, Dongwon Kim, HyungRok Jung, **Yoonjae Baek**, Byung-kwan Lee, Suha Kwak, Jeany Son
Under Review, 2026

[3] ***EventCoT: Event-centric Video Chain-of-thought for Reasoning Temporal Localization***
Youngkil Song, **Yoonjae Baek**, Dongwon Kim, Inho Kim, Dongkeun Kim, Suha Kwak
Under Review, 2026

[4] ***Dual Path Semantic Segmentation with Locality Guidance Tokens***
Jaeik Kim, **Yoonjae Baek**, Aecheon Jung, Hongyeob Kim, Soyun Choi, Youngeun Kim, Jaeyoung Do, Sungeun Hong
Neurocomputing, 2026

Experience

Computer Vision Lab, POSTECH — Pohang, South Korea
Research and Teaching Assistant — *Feb 2025 – present*

AI & Media (AIM) Lab, Sungkyunkwan University — Seoul, South Korea
Undergraduate Research Intern — *Jan 2024 – June 2024*
- Advisor: Prof. Sungeun Hong

Honors and Awards

Gold Prize, 38th Workshop of Image Processing and Image Understanding (IPIU), top 3 of 368 papers, 2026

Services

Student Volunteer, Conference on Robot Learning (CoRL 2025), Seoul, South Korea, 2025

Leadership

President, Student Council of the Department of Computer Education, Sungkyunkwan University, 2019–2020

Interests

Embodied AI, Vision-Language-Action Models, Real-World Deployment, Video Understanding